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Compounding preconditions of wildfires vary in time and space within Europe

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Favorable wildfire conditions are increasing in frequency and severity across Europe. Understanding how wildfire drivers vary in space and time is crucial for mitigating wildfire risk under current and future climate conditions. Here, we analyze the hydro-meteorological and land-surface drivers of wildfires from 2001 to 2020 across eight European climate regions and their mountain ranges. Our findings reveal that drought conditions and vapor pressure deficit are the dominant drivers of wildfire activity. These drivers vary by season and region: in Southern and Central European regions, persistent warm and dry conditions in preceding seasons favor summer wildfires, while fall wildfires are influenced by fuel build up in spring that loses moisture during dry and hot summer weather. In Northern Europe, these dynamics occur on sub-monthly timescales. Our results illustrate the critical role of compounding wildfire drivers and emphasize the need for targeted mitigation strategies, especially in the light of climate change.

The wildfire seasons of 2023 and 2022 were ranked among the five worst wildfire seasons of the European continent since the beginning of the century[1,2]. Even though burned area shows a long-term decline in the Mediterranean region[3,4] and at the global scale[5], burned area increases in mid- and high-latitude climate regions[4]. In these regions, increasing wildfire activity is especially concerning, because of abundant fuel availability[6], and lower adaptation and resilience of plants to fires[7].

In recent years, Central and Northern Europe have experienced severe wildfires, promoted by intense drought conditions that led to high dry fuel availability in a usually moisture-limited fire regime[8], e.g., in Northern Europe during the drought in 2018[9,10] and in Central Europe during the drought of 2022[2,11]. Under climate change, warmer and drier climate zones are expected to expand towards northern European latitudes[12,13], enhancing fire weather and dry fuel availability, which consequently increases the risk of large wildfires in these regions[14–16]. As wildfire-favoring conditions intensify in historically less wildfire-prone areas, and wildfire seasons increase in length in already wildfire-prone regions of Europe[17], a more detailed understanding of the individual and compounding wildfire preconditions, and their temporal persistence across seasons and climate regions of Europe is needed.

Wildfires can be understood as compound events that result from spatially and temporally compounding hydro-meteorological and land-surface conditions that favor wildfire ignition and spread[18]. These conditions include both meteorological conditions, such as high temperatures, strong vapor pressure deficits (VPDs), and high wind speeds[4], as well as land-surface conditions such as drought conditions resulting from precipitation and soil moisture deficits, which modulate fuel loads and availability[4,8]. Although there is a strong link between meteorological fire weather described by fire weather indices, such as the Canadian Forest Fire Weather Index (FWI)[19], and wildfire occurrence[20,21], fuel loads and availability are not explicitly represented in these index systems[22]. Still, the FWI accounts for fuel moisture in its fuel moisture codes, which empirically capture drought conditions in different fuel layers for up to 52 days[19]. We hypothesize that wildfire preconditions vary by season and region, and that understanding these characteristics requires analyzing individual hydro-meteorological and land-surface variables, such as gross primary productivity, soil moisture, and snow depth, rather than solely relying on compound indices like the FWI. Therefore, we here examine each hydro-meteorological and land-surface variable separately, without assuming any a priori compounding relationship between the individual wildfire drivers.

The amount and timing of biomass accumulation are crucial for wildfire activity as gross primary productivity in wet periods modulates fuel availability for the following dry periods, e.g., the growth of Mediterranean shrubs in wet periods in Mediterranean climates[8,23]. In regions where fire activity is moisture-limited, prolonged dry periods decrease fuel moisture and the risk of large wildfires[24,25]. The processes of fuel accumulation and drying act—depending on the ecosystem—on seasonal to annual timescales[23,26]. In contrast, meteorologically distinct fire weather is characterized by hot, dry, and windy atmospheric conditions that occur on daily

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to weekly timescales and are influenced by large-scale weather patterns over Europe[27,28].

Fire weather, season length, and vegetation dynamics vary across environmental zones[29], within which climate variability modulates annual burned area fluctuations[30]: In Mediterranean and more arid climates, pronounced antecedent wet periods can promote above-average biomass productivity, leading to higher fuel loads in subsequent dry periods that can result in extreme wildfire activity[30–33], e.g., as observed in Portugal in 2017[26]. In contrast, in temperate and boreal climate zones, where wildfire occurrence is moisture-limited, sub-seasonal drought conditions increase dry fuel availability, which in turn promotes higher wildfire activity than average climate conditions[30,34], e.g., as in Sweden in 2018[35]. In mountain regions, such as the southern Alps, prolonged dry periods and snow-scarce conditions in winter, in combination with strong katabatic winds, i.e., Foehn winds, promote snow-free fuel layers resulting in fast-spreading winter wildfires[24,36]. These examples highlight that inter-annual variability affects wildfire preconditions differently in each region and season. Therefore, a detailed analysis of climate–vegetation–wildfire interactions is needed to understand region- and season-specific wildfire danger.

Until now, wildfires in Europe have been mostly studied on national and regional levels, with a focus on the highly wildfire-affected areas in Southern Europe. Recent severe wildfire seasons that coincided with strong drought conditions in Central and Northern Europe suggest that the "switch"[8] for flammable conditions in moisture-limited regions is turned on. Given that wildfires and their severity are not exclusively modulated by short-term weather conditions, but also by antecedent wet-dry transitions, wildfire occurrences and their preconditions need to be studied from a holistic perspective. Such a holistic perspective should include land-surface parameters and hydro-meteorological drivers that go beyond traditional fire weather indices and consider daily, seasonal, and annual timescales. In this study, we highlight the importance of individual wildfire drivers for the occurrence of wildfire events in different European regions by using random forest models and assess when preconditions start becoming anomalous before a wildfire event. We analyze the three-month Standardized Precipitation Evaporation Index (SPEI-3M), soil moisture, and snow depth deficit as long-term wildfire drivers and temperature, VPD, and wind speed as short-term wildfire drivers. In addition, we consider gross primary productivity (GPP) (accumulated over 1 month) as a proxy for fuel drying, which is influenced by both, short- and long-term drivers. By analyzing all of these potential wildfire drivers individually, we are able to disentangle the compounding preconditions that lead to wildfire occurrence in Europe. To address the challenge of wildfire drivers acting on different timescales, we propose the concept of "time of precondition emergence" (ToPE). This concept will allow us to better understand the relative importance of and interplay between long- and short-term wildfire drivers, as well as the temporal onset of wildfire preconditions. Such understanding is highly relevant for early warning and wildfire prevention, especially under changing climate conditions.

Here, we disentangle the regional and temporal characteristics of hydro-meteorological and land-surface drivers of wildfires in Europe. We divide Europe into eight different climate regions, namely the PRUDENCE regions[37], and account for altitude-related climate gradients within these regions by distinguishing between high-mountain and non-mountain regions[38]. We study the regional and seasonal characteristics of wildfires and their drivers by (i) describing the seasonal and regional characteristics of wildfire occurrence in Europe, (ii) comparing the hydro-meteorological and land-surface conditions of days when wildfires start compared with days when no wildfires are burning (non-wildfire days), (iii) understanding to which degree these conditions differ between high-mountain and non-mountain regions, (iv) investigating regional differences in the most important drivers of wildfires, and (v) quantifying the length of the time period during which hydro-meteorological and land-surface conditions have been anomalous prior to wildfire events.

Results and discussion

Seasonal and regional characteristics of wildfires

We study the spatial and temporal patterns of wildfires across eight regions [i.e., PRUDENCE regions[37]] and their respective high mountain ranges [i.e., high and scattered high mountains in ref. 38] by leveraging the ESA FireCCI version 5.1 (FireCCI51) burned area pixel product[39] between 2001 and 2020. We assess the spatial distribution of wildfires by the cumulative burned area over the 20-year study period and describe the seasonality in burned area and number of wildfire events as each season's fraction of the regional and mountain range types of the 20-year total.

Wildfire occurrence and peak seasonality in Europe vary greatly across the eight examined regions and their mountain ranges (Fig. 1). Wildfires are most common in summer in Southern Europe (Iberian Peninsula and Mediterranean), France, the mountain regions of Eastern Europe and Scandinavia, and in the non-mountain parts of the Alps. Regions characterized by summer wildfires experience wildfires in fall as well, though the seasonality is less pronounced than in summer. In the fall, mountain regions are more strongly affected by wildfires than non-mountain regions, whereas in summer, non-mountain regions are more strongly affected. In the British Isles, the non-mountain regions of Scandinavia, Mid-Europe, and Eastern Europe, wildfire seasonality peaks in spring.

In terms of total burned area over the 20-year observation period (i.e., 2001–2020), regions in Southern Europe are the most strongly affected by wildfires, with large burned area clusters in the western Iberian Peninsula, southern Italy, and the Balkan countries (see Fig. 1). The second region most strongly affected by wildfires is Eastern Europe, where wildfires occur along the Carpathian mountain range, but are most dominant on the border between Ukraine and Belarus, as well as the Russian exclave Kaliningrad. In Scandinavia, wildfires occur mostly in the non-mountain regions of Sweden and the Baltic states (i.e., Estonia, Lithuania, and Latvia), while mountain regions experience fewer wildfires. In contrast, the mountain ranges of the British Isles, including the Scottish Highlands, show frequent wildfire occurrence. Mid-Europe, France, and the Alps are the regions with the fewest observed wildfires, but each of these regions has at least one wildfire hotspot, i.e., the Atlantic Coast and Massif Central in France, the southern mountain ranges in the Alps, and the northern parts of Mid-Europe.

Hydro-meteorological and land-surface conditions of wildfires

We analyze hydro-meteorological and land-surface conditions of wildfire events in terms of seasonal and absolute anomalies in the eight PRUDENCE regions, subdivided into their mountain and non-mountain areas. We describe the conditions of wildfire events on their first day of detection, i.e., wildfire start days, and compare these conditions to days, when no wildfires are detected, i.e., non-wildfire days, using our event detection algorithm (see "FireCCI51 and derived wildfire events" under the "Methods" section). We exclude conditions on days that are not wildfire start days to avoid oversampling the same event in case it burns for multiple days. For each wildfire start day and non-wildfire day, we sample short-term drivers, i.e., maximum temperature, vapor pressure deficit (VPD), 10-m surface wind speed, and long-term drivers, i.e., 3-month Standardized Precipitation Evapotranspiration Index (SPEI-3M), soil moisture, and snow depth deficit, as well as gross primary productivity accumulated over 32 days (GPP-1M) as a proxy for fuel drying. We apply a Mann–Whitney *U*-test to test whether there is a significant difference between wildfire start and non-wildfire days, and between mountain and non-mountain regions.

Wildfires occur in all seasons and regions under warmer and drier conditions than the seasonal and regional mean (Fig. 2). These seasonal hot and dry conditions likely affect GPP-1M, which reflects the availability of dry fuels, and hence affects region-specific wildfire activity (see Fig. 2). We find that in spring and summer, GPP-1M is less abundant on wildfire start days than on non-wildfire days in all regions except Scandinavia and the British Isles, where GPP-1M is more abundant on wildfire start days in

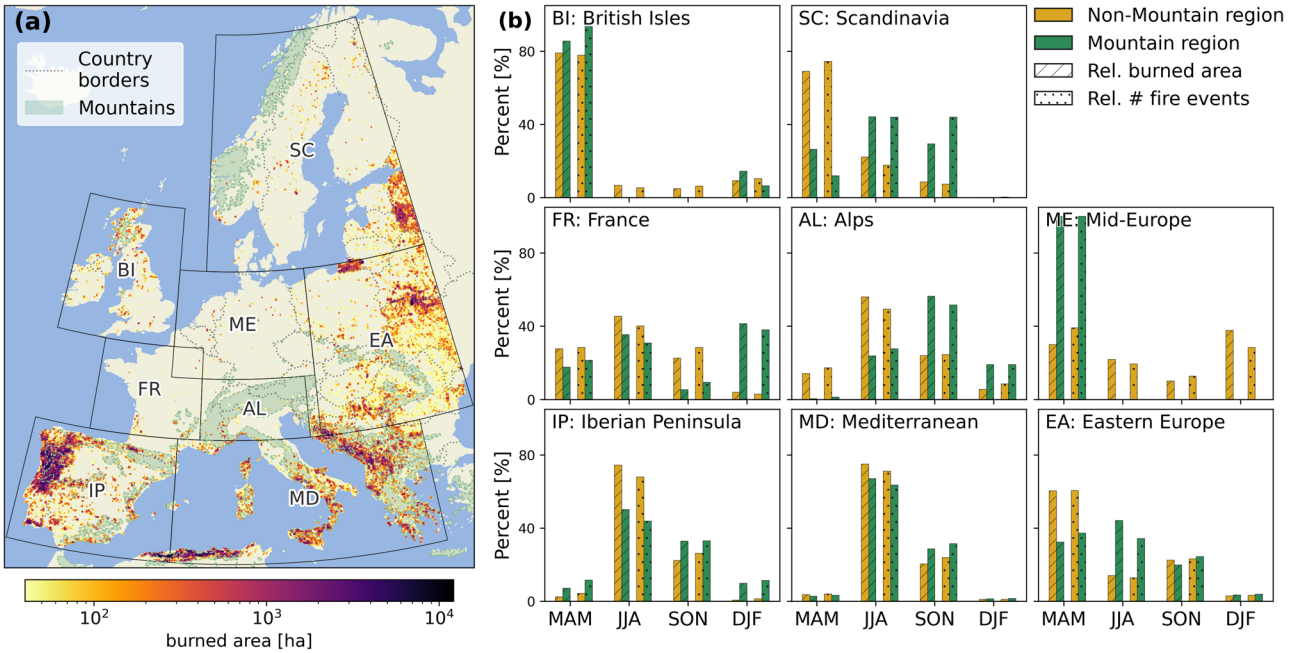


Fig. 1 | Spatial and temporal distribution of wildfires in Europe. a Cumulative burned area (2001–2020) per CERRA grid-cell (5.5 km) for Europe (BI British Isles, FR France, IP Iberian Peninsula, SC Scandinavia, ME Mid-Europe, AL Alps, MD Mediterranean, EA Eastern Europe). **b** Relative burned area (hatched) and number of wildfire events (dotted) of the annual mean sum derived for each region and its respective mountain (green) and non-mountain regions (yellow).

comparison to non-wildfire days. In fall and winter, GPP-1M is more abundant in all regions on wildfire start days than on non-wildfire days. Seasonal temperature anomalies are less strongly pronounced than SPEI-3M, soil moisture and VPD anomalies (see Fig. 2). In summer, wildfire drivers on wildfire start days are less strongly pronounced in Southern Europe in comparison to other regions, which suggests that wildfire-favoring conditions might be met on non-wildfire days too (see Supplementary Material Fig. S2), or wildfire activity is related to other factors, which are not considered in this study. Seasonal wind speed is higher on wildfire start days in comparison to non-wildfire days in Southern Europe (Iberian Peninsula and Mediterranean) in summer, in fall in the British Isles and Alps, and in winter in the Alps. In contrast, wind speed in the remaining regions and seasons is lower on wildfire start days in comparison to non-wildfire days. We observe a large variability in wind speed on wildfire start days (see Supplementary Material Fig. S2) and assume that the 5.5 km spatial resolution limitedly resolves the 10-m daily surface maximum wind speed during initial wildfire ignition. Seasonal snow depth deficits are pronounced (i.e., ≥0.25 SD) on wildfire start days in winter in Northern and Eastern Europe and in France, the Alps, and Mid-Europe. In spring, the seasonal snow depth deficits on wildfire days are less strongly pronounced than in winter, and are only larger than +0.25 SD in the Alps and Scandinavia. While wildfires under these snow depth deficits most likely occur in snow-free periods, such snow-free periods are atypical in these regions and seasons, as shown by the comparison to non-wildfire days.

Although the standardized seasonal anomalies of the examined wildfire drivers are distinct, looking at these drivers from an absolute perspective provides further insights: Absolute temperature and VPD in summer are high in Southern Europe on wildfire start days (see Fig. S2). However, we find that these variables are, from a seasonal perspective, not strongly different from those on non-wildfire days (see Fig. 2). Strong seasonal soil moisture deficits on wildfire start days reflect an absolute deficit of more than -0.5 SD in spring, summer, and fall. In contrast, seasonal soil moisture anomalies in winter originate from an absolute soil moisture surplus, which is still lower than on non-wildfire days (see Fig. S2). The SPEI-3M is less season-dependent, and the identified seasonal anomalies are of the same magnitude as the absolute anomalies (see Fig. S2).

Different wildfire conditions in mountain vs non-mountain regions

We find that most drivers of wildfires are more pronounced in mountain regions than in non-mountain regions (see Fig. 2), which indicates that wildfires in mountain regions start under seasonally stronger anomalies than wildfires in non-mountain regions. However, these differences are significant only in a few regions and seasons: On wildfire days we find that wind speed, snow depth deficit (e.g., Northern Europe in spring and the Alps in fall and winter) and SPEI-3M (e.g., Eastern Europe in summer) are significantly enhanced in mountain regions, when compared to non-mountain regions (p-value ≤ 0.05) (see Fig. 2, "On wildfire start days" panel). On wildfire start days, maximum temperature (e.g., British Isles and Alps in summer), GPP-1M (e.g., in Mid-Europe and Eastern Europe in spring and in Eastern and Southern Europe in winter), and SPEI-3M (e.g., in Scandinavia in summer and fall) show stronger anomalies for wildfire-favoring conditions in non-mountain than in mountain regions. These anomalies reflect the elevational and seasonal dependencies of wildfire drivers in the respective regions and seasons.

Most important wildfire drivers by region and season

We identify the most important drivers of wildfire events on their start day for each region and season by utilizing the variable importance metric of random forest models that predict binary wildfire occurrence (see details in the chapter "Variable importance of wildfire drivers" under the "Methods" section). The overall accuracy of the confusion matrix of the random forest models, which were trained on a balanced wildfire start day vs non-wildfire day dataset, ranges between 0.76 and 0.96 (overall accuracy, see Supplementary Materials Fig. S3) and provides a good baseline to interpret variable importance derived from the Gini-Impurity index.

The overall dominant drivers of wildfires are SPEI-3M, soil moisture deficit, and VPD. However, the relative importance of these wildfire drivers varies regionally and seasonally as they interact with other wildfire drivers, such as maximum temperature and GPP-1M. In spring and summer, the most important drivers of wildfire events are GPP-1M and VPD, while SPEI-3M and VPD are crucial in fall and winter (see Fig. 3). In spring, GPP-1M is the most important wildfire driver in all regions, except in the British Isles, France (ranked 2nd), and the Alps (ranked 3rd). In spring, the

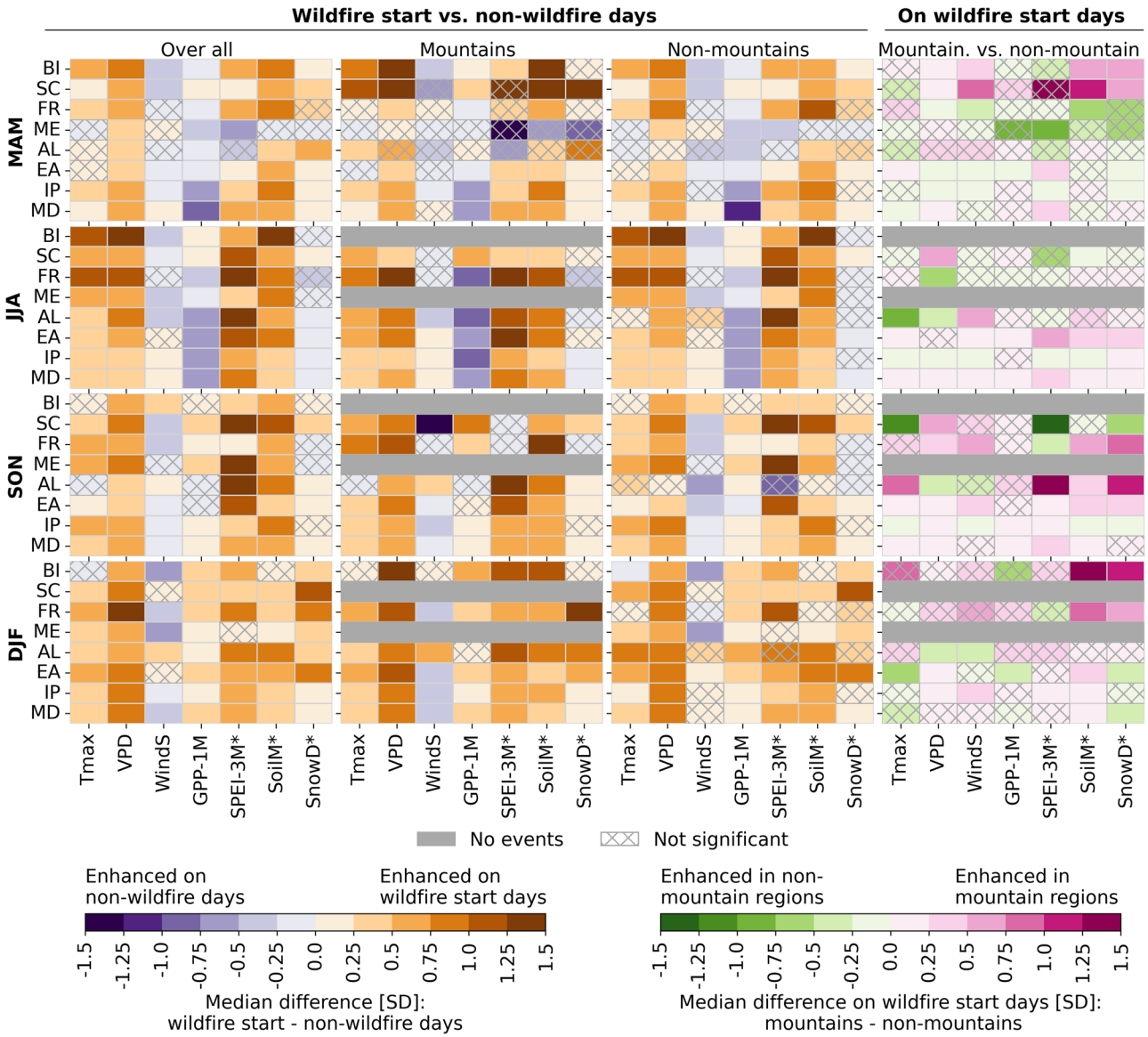


Fig. 2 | Standardized anomalies of hydro-meteorological and land-surface drivers on wildfire start days in comparison to non-wildfire days. We show 2-m maximum temperature (Tmax), 2-m vapor pressure deficit (VPD), 10-m surface wind speed (WindS), 32-day accumulated (GPP-1M), 3-month SPEI (SPEI-3M), soil moisture (SoilM), and snow depth (SnowD) on the start day of wildfires in comparison to days without wildfires for each region and season over the entire subregion ("Over all" panel; left), mountain regions ("Mountains panel"; middle left) and non-mountain regions ("Non-mountains"; middle right). Colored values are based on the seasonal median difference between all wildfire start events and all non-wildfire events. The right panel ("On wildfire start days") shows the difference between mountain regions and non-mountain regions on wildfire start days. Hatching indicates that the difference between wildfire start and non-wildfire days (first three panels from left) and mountain vs non-mountain regions (right panel) is not significant (Mann–Whitney *U*-test, p-value ≤ 0.05). Gray panels indicate that no events are observed in this region and season. To have a uniform sign across all variables, we flip the sign for deficit variables and mark these with an asterisk (*), i.e., SPEI-3M soil moisture and snow depth.

importance of GPP-1M as a wildfire driver reflects a seasonal deficit of GPP-1M in all regions (see Fig. 2). VPD and soil moisture deficit are the second and third most important variables in spring, as they are ranked among the three most important drivers in all regions. In summer, the most important drivers differ between the northwestern (i.e., France, Mid-Europe, British Isles, and Scandinavia) and southeastern (i.e., Iberian Peninsula, Alps, Mediterranean, and Eastern Europe) regions: in the southeastern regions, seasonal GPP-1M deficit is still the most important variable together with SPEI-3M and VPD. In the northwestern regions, VPD, maximum temperature, and a long-term drought variable (either SPEI-3M or soil moisture deficit), represent the most important wildfire drivers (see Fig. 3). In fall, long-term drought conditions, as described by the SPEI-3M and soil moisture deficit, along with VPD, are the most important variables for wildfire occurrence. In Southern Europe and Mid-Europe, elevated temperatures play an important role, too, while in the remaining regions, temperature is not ranked among the three most important variables. In winter, wildfires are most importantly influenced by VPD. Besides high VPD, long-term drought conditions reflected in the SPEI-3M are ranked among the three most important drivers of wildfires in the Iberian Peninsula, the British Isles, France, and the Alps. In addition, GPP-1M surpluses become crucial for wildfire occurrence in Southern Europe, France, and the British Isles. Positive temperature anomalies play a critical role in winter wildfire occurrence in Scandinavia, Mid-Europe, Eastern Europe, and the Mediterranean.

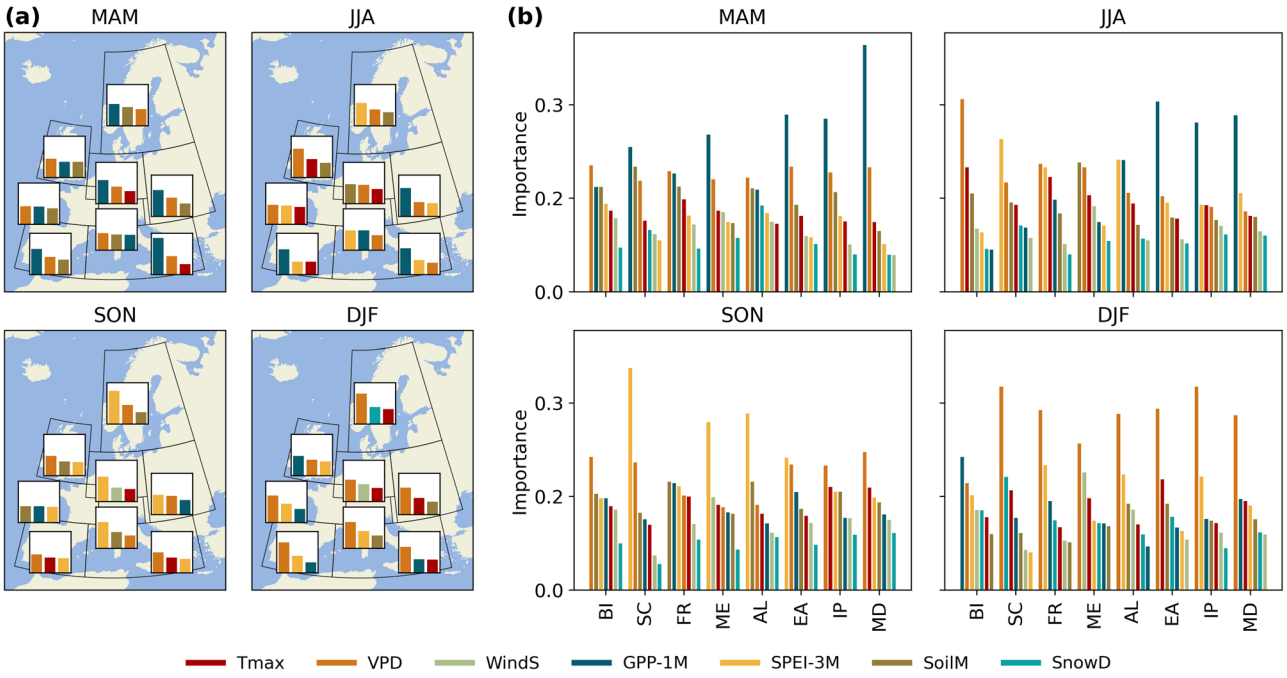


Fig. 3 | Drivers of wildfires ranked by random forest feature importance. a Three most important variables on the start day of wildfire events based on the Gini impurity index derived from seasonal and regional random forest models. **b** Variable importance by region and season for all drivers used in the random forest models.

Time of precondition emergence

We investigate how long individual wildfire drivers have been anomalous before the start of wildfires. Specifically, we average the absolute anomalies of each driver over incrementally increasing time steps, to identify the time window during which conditions on days when wildfires start differ notably from conditions on non-wildfire days. The time steps considered range from the day of wildfire start, i.e., day 0, up to one year before the day of wildfire start, i.e., 360 days. We call the time step when wildfire start day anomalies emerge from 1 standard deviation (SD) of the difference between wildfire start day and non-wildfire day anomalies the "time of precondition emergence" (ToPE). This 1 SD of the difference is mapped onto the non-wildfire conditions to account for natural variability (see Fig. 4a). For example, in summer in the Mediterranean region, maximum temperature and VPD and SPEI-3M anomalies emerge 270 and 240 days before wildfire events start from non-wildfire conditions, respectively, while GPP-1M deficits emerge 60 days before wildfire events start (see Fig. 4b). The long ToPE times for maximum temperature, VPD and SPEI-3M, imply that preceding months have been anomalously warm, atmospherically dry, and precipitation-scarce. In summary, these findings show how the preconditions of wildfires develop over time. The example of Mediterranean summer shows how preceding warm and dry spring and winter months lead to the availability of dry fuels, i.e., a GPP-1M deficit exceeding 1 SD 60 days prior to wildfire start days (see Fig. 4b).

By looking at all ToPE times for each variable, season and region, we find strong seasonal and regional patterns, with generally shorter ToPE times in spring and winter, and generally longer ToPE times in summer and fall (see Fig. 4c): In spring in Southern (Iberian Peninsula and Mediterranean) and Eastern Europe, SPEI-3M emerges from non-wildfire conditions more than 270 days before wildfire events occur. These drought conditions likely lead to a GPP-1M deficit that emerges 90 days before wildfire events start. VPD and snow-depth deficit emerge on a sub-monthly scale (ToPE ≤ 30 days) in these regions. In Northern Europe (Scandinavia and British Isles) and France, wildfire preconditions emerge up to 14 days prior to wildfire events (see Fig. 4c). In summer, we observe persistent wildfire preconditions longer than 90 days in all regions, except Mid-Europe and the British Isles. These persistent preconditions of high maximum temperatures and VPD, alongside pronounced soil moisture deficits and low SPEI-3M, likely lead to GPP-1M deficits, that develop 60 to 30 days prior to wildfires. ToPE times in fall are shorter in Southern Europe and France, longer in Eastern Europe, the Alps, and Mid-Europe, and equally long in Northern Europe in comparison to the summer season. We find that SPEI-3M is anomalously low in Mid-Europe, the Alps, and Eastern Europe a year before wildfire starts in fall, and likely leads to soil moisture deficits that emerge on shorter timescales than temperature and VPD. GPP-1M surpluses for fall wildfires emerge from non-wildfire conditions in the previous vegetation period, i.e., ToPE ≥ 90 days in France, the Iberian Peninsula, and Scandinavia. We show that for these regions, wildfires in fall have relatively short ToPE times for VPD, temperature, and soil moisture compared to ToPE for GPP-1M surpluses, leading to dry fuel availability. ToPE times ≥ 180 days for snow depth deficits in the Alps and the Mediterranean region, alongside of persistent (ToPE times ≥ 90 days) SPEI-3M conditions show that fall wildfires can be influenced by the previous season's snow cover (see Fig. 4c). For winter wildfires, snow depth deficits develop on same-season timescales (ToPE ≤ 90 days) in the British Isles, Scandinavia, the Alps, France, Eastern Europe, and the Iberian Peninsula. In these regions, SPEI-3M is anomalously low for longer periods than snow depth deficits, and VPD and temperature are anomalously high on shorter timescales than snow depth deficits. We find that GPP-1M surpluses in winter are anomalously high from the previous active vegetation period, with ToPE times ≥ 240 days in France and Southern Europe. These findings illustrate how long-term drought conditions in combination with anomalously high VPD can promote wildfire danger in winter.

Wildfires result from compounding drivers

Our results highlight how land-atmosphere interactions affect wildfire occurrence by illustrating how hydro-meteorological drivers and drought conditions affect fuel availability, i.e., available biomass to burn, approximated from GPP. We find that pronounced dry conditions, resulting from short-term atmospheric dryness (i.e., VPD) and long-term drought (i.e., SPEI-3M or soil moisture deficit), are the most important drivers of wildfire occurrence in Europe, whereas temperature and wind speed play a secondary role (see Fig. 2). We find that the seasonal anomalies for all variables vary mostly in the range of 0.25 and 1 SD and single variable outliers larger than 1 SD occur mostly in summer and fall. These findings highlight that the

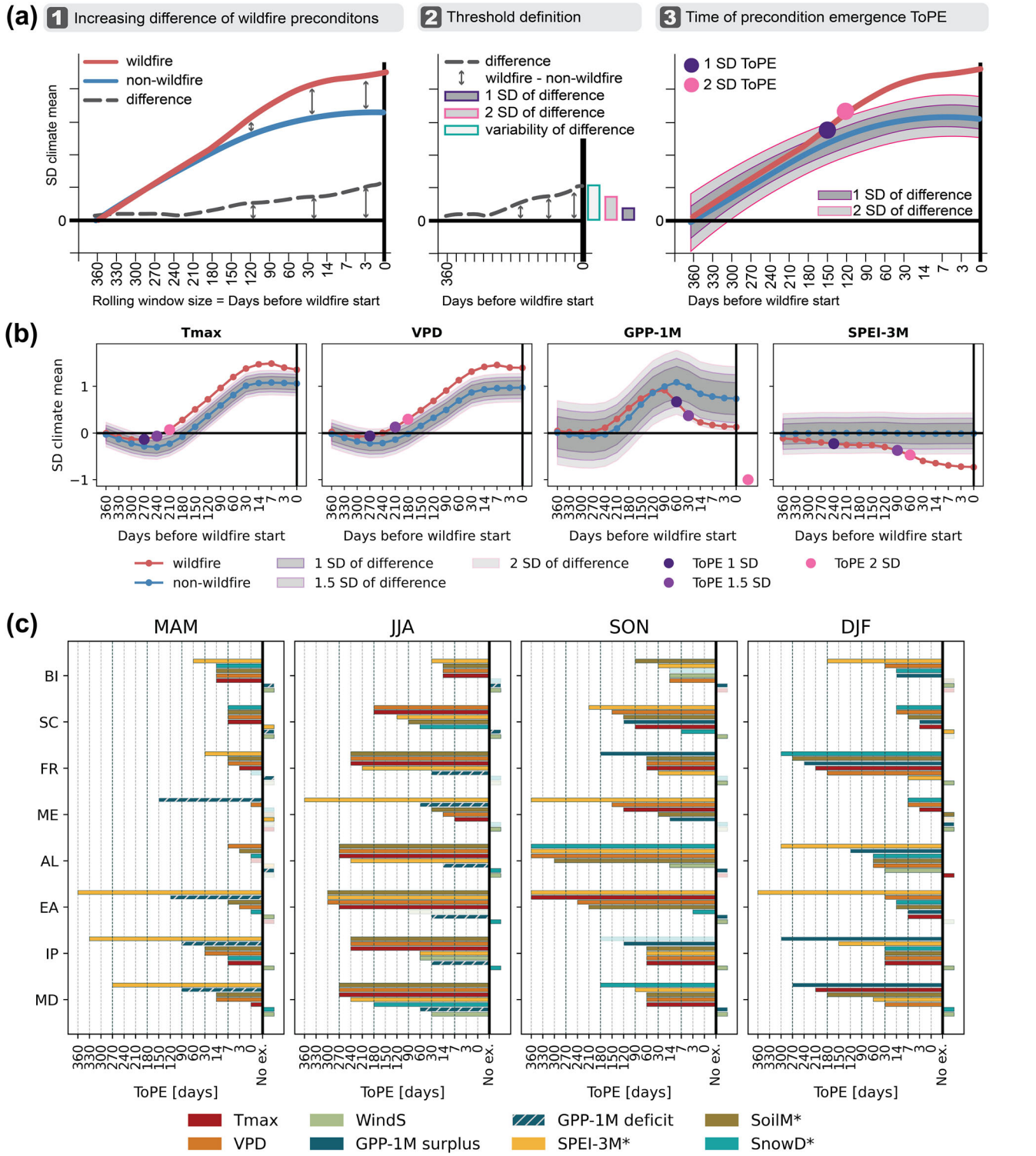


Fig. 4 | Time of precondition emergence (ToPE) for wildfire drivers: framework and results. a Conceptual framework developed to determine ToPE, illustrating how emergence levels of all driving variables are derived for a threshold of 1 and 2 SDs of the difference between wildfire and non-wildfire conditions. **b** Example of how ToPE is derived for different variables for the Mediterranean region in summer (JJA). If ToPE does not occur at the 1 or 2 SD threshold level, the dot is plotted at the bottom-right of the individual subplot (e.g., 2 SD for GPP-1M). **c** ToPE for all variables exceeding the threshold of 1 SD, which is derived from the difference between wildfire and non-wildfire values for the respective variable (color), season (panel), and region (*y*-axis). Variables with an asterisk (*) in the legend are deficit variables, hatched GPP-1M values represent deficits, and solid-filled GPP-1M values represent surpluses. Faded variables are not significantly different (p-value ≤ 0.05) on wildfire start days compared to non-wildfire days according to the Mann–Whitney U-test.

co-occurrence of multiple moderately pronounced drivers is more relevant for wildfire activity than strong anomalies in one variable alone, which is in line with findings for other regions of the world, such as California[40].

In Europe, the development of cascading wildfire preconditions starts with biomass drying (i.e., GPP-1M deficit) as biomass is sufficiently available even in the driest climate regions of Europe, i.e., Southern Europe[8,23]. The availability of dry fuel results from interactions between GPP-1M, temperature, and dry conditions, i.e., soil moisture deficits, low SPEI-3M, and VPDs (see Fig. 4). Wildfire occurrence is restricted by fuel availability in dry, or "fuel-limited", regions such as the driest parts of Southern Europe, and is "moisture-limited" in regions with high GPP but sufficient moisture, such as Northern and Central Europe[8]. Our results of these interactions represent relative changes in the regional and seasonal live fuel moisture levels rather than the full region-specific fuel load and structure as in other studies that use a sophisticated fuel model [i.e., refs. 31,41,42]. Still, we illustrate how hydro-meteorological conditions influence the flammability of biomass (see Fig. 2): in spring and summer, GPP-1M is lower on wildfire start days than under non-wildfire conditions, indicating that VPD and soil moisture deficits likely lead to a decrease in GPP-1M and promote its drying relative to non-wildfire conditions. However, in Northern Europe (Scandinavia and the British Isles), high VPD, low SPEI-3M, and soil moisture deficits are associated with slightly increased (≤ 0.25 SD) GPP-1M on days when wildfires occur in summer. This suggests that GPP-1M decreases only when soil moisture is below a certain absolute threshold[43]. With respect to wildfire occurrence, our findings imply that GPP-1M deficits in spring and summer represent decreases in fuel moisture in Southern and Central Europe, while GPP-1M surpluses in Northern Europe under drought conditions[44] become rapidly available as dry fuel under high VPD and high temperatures (see Figs. 2 and 4).

Our results for the ToPE highlight memory effects of all driving variables from daily to annual timescales prior to wildfire events and illustrate the temporal development of compounding preconditions for wildfires in different seasons and regions (see Fig. 4). While regional differences are present, we identify distinct propagation patterns of wildfire preconditions in different seasons: In spring, ToPE times are short, suggesting that wildfire preconditions develop rapidly in this season. In Southern and Eastern Europe, preceding dry winter and fall conditions (SPEI-3M) additionally favor wildfires in spring (see Fig. 4c). In summer, ToPE times for maximum temperature, VPD, SPEI-3M, and soil moisture deficit are longer than 90 days in Southern Europe, Eastern Europe and the Alps showing that wildfire preconditions manifest early in the fire season, while wildfire preconditions develop on shorter timescales in the remaining regions (see Fig. 4c). For wildfires in fall, we demonstrate that fuel (i.e., GPP-1M surplus) is accumulated in spring (Mediterranean, Iberian Peninsula and France) and summer (Scandinavia), which likely becomes available as dry fuel under high temperatures and enhanced VPD, which start developing up to 30 days before wildfires start. For these regions, ToPE patterns are similar in the winter, however, ToPE is comparatively longer for GPP-1M and shorter for temperature, VPD, and soil moisture. In winter in Northern Europe (British Isles, Scandinavia), Eastern Europe, and the Alps, long-term drought conditions (SPEI-3M) lead to snow depth deficits that promote wildfires under high VPD. ToPE returns the longest time step prior to an extreme event for which a specific precondition variable starts to emerge from non-extreme event conditions. This time step does not necessarily represent the peak of the precondition anomaly, rather, it indicates its earliest onset, and, therefore, the start of the cascade of anomalous preconditions for wildfires. Our ToPE analysis provides insights into how quickly dangerous wildfire preconditions develop in different regions and seasons, which can inform wildfire forecasting and management.

The role of wind and snow in wildfire occurrence

According to our analysis, the most important drivers of wildfire occurrence are long-term drought conditions and VPD throughout all seasons, while wind speed and snow depth are only tangentially important in most regions (see Figs. 3 and 2). For most regions and seasons, we find that wind speed is significantly lower on wildfire start than on non-wildfire days, except in Southern Europe (Iberian Peninsula and Mediterranean) in summer, the British Isles and the Alps in fall, and the Alps in winter (see Fig. 2 panel "Overall"). In the mountain regions of the Alps, wind speed is particularly relevant, because wind-driven wildfires in winter co-occurring with snow depth deficits explain the majority of the total annual burned area in this region[24]. Nevertheless, wind speed is a driver of wildfire size in other regions and seasons as well[45], especially in regions with local wind systems, such as the "Mistral" in France[46] or the "Melteni" in Greece[4,47]. Our results show a high variability of wind speed on wildfire start days (see Supplementary Material Fig. S2), suggesting that daily 10-m maximum surface wind speed in the 5.5 km CERRA grid is not sufficient to resolve the high sub-daily and spatial variability of wind speed on wildfire start days, and that wind speed might be more relevant for wildfire spread than ignition.

Though the majority of wildfires occur in summer and fall across Europe, wildfires in spring and winter represent a substantial fraction of all wildfire events in Scandinavia, British Isles, Eastern Europe, France and the Alps (see Fig. 1b). Winter and spring wildfires in these regions are accompanied by snow depth deficits, high VPD and soil moisture deficits (see Fig. 2). In these snow-dominated regions, snow depth deficits likely lead to snow free periods, that create potential for wildfire activity by revealing dry surface litter from previous seasons[36,48]. In the period between snow melt and the onset of photosynthesis in spring, often referred to as a "spring dip"[48,49], surface litter is highly flammable, leading to a peak in wildfire activity across boreal biomes[48]. This phenomenon has mostly been studied in North America[48–50] and, to a lesser extent, in the European Alps[24,36]. However, our findings suggest that the "spring dip" and its underlying conditions are also present in snow-dominated regions across Europe, i.e., Scandinavia, the British Isles, Eastern Europe, France, and the Alps (see Figs. 2 and 3).

Methodological limitations and uncertainties

In our study, we used the ESA FireCCI51 product to identify wildfire events and to robustly distinguish between the hydro-meteorological and land-surface conditions on wildfire start days vs days when no wildfires are burning (non-wildfire days). However, satellite-derived wildfire observations come with some limitations and uncertainties: First, wildfire observations from space, such as the ESA FireCCI51 product, have high detection uncertainties for small wildfires[51,52]. To exclude false detections, we only analyzed wildfires that are larger than 40 ha and have a detection confidence larger than 0.6. We acknowledge that these thresholds might have limited the sample size of wildfire events in regions with smaller wildfires, and we point out that our results related to wildfire drivers only apply to larger wildfires, i.e., those ≥40 ha. It remains to be analyzed if the importance of wildfire drivers changes with decreasing wildfire size. Second, we observe distinctive discontinuities in cumulative burned area along several country borders, e.g., Belarus and Ukraine, Russia (i.e., Kaliningrad), in the Alps on the Italian border of France and in the Pyrenees on the French border of Spain (see Fig. 1). These differences in wildfire event properties most likely originate from differences in wildfire management, forest structure and land ownership, which differ across national or even sub-national boundaries[53–55]. Third, satellite-derived wildfire products cannot account for the cause of fires, and therefore do not differentiate between prescribed burns vs wildfires. Distinguishing between human and lightning-ignited wildfires would enable accounting for the seasonality of lightning strikes and show that hydro-meteorological preconditions needed for wildfires differ by their ignition source[56].

Fourth, heat sources from volcanoes and gas flaring can lead to active wildfire detections in satellite-based wildfire products[57]. However, because the ESA FireCCI51 product incorporates a hybrid approach by considering vegetation changes in their burned area detection[39], and we filtered for wildfire detections on natural land cover types only (see "FireCCI51 and derived wildfire events" under the "Methods" section), we are confident that our wildfire event dataset only includes a small number of wrongly detected wildfire events.

In this study, we used random forest models and their feature importance output to understand drivers of wildfire events. As feature importance estimates can be biased when predictor variables are correlated[58], we excluded highly correlated variables from the random forest model setup by using VPD and dropping relative humidity. For the variables used in the random forest model setup, we find that multiple variables have a strong correlation with temperature (see Supplementary Material Fig. S5): VPD and soil moisture have the strongest correlation with temperature, but also GPP-1M in spring and fall correlate strongly with temperature and VPD (i.e., $R^2 \geq 0.5$, see Supplementary Material Fig. S5). VPD and SPEI-3M are compound variables that, by definition, depend on temperature and therefore might better predict burned area than non-compound variables. The high correlation of multiple variables with temperature explains the lower ranking of temperature as a driver of wildfire events in comparison to other drivers. However, Fig. 2 shows that VPD and drought-related variables (i.e., SPEI-3M, soil moisture) are more strongly anomalous on wildfire start-days in comparison to non-wildfire days than temperature. Therefore, the correlation of these variables with temperature might slightly influence the results of the random forest analysis. However, the core findings of our analysis—that long-term drought conditions and VPD are the most important drivers of wildfires—are robust and consistent with results of other studies focusing on wildfire drivers [e.g., refs. 23,34,59,60].

Conclusion

We observe wildfire-favoring conditions not only in summer and in Southern Europe, which has been the focus of many previous studies, but also in all other seasons and regions of Europe. While long-term dry conditions, described by the SPEI-3M and soil moisture deficit, and atmospheric dryness, described by VPD, are the strongest and most important drivers of wildfires, across seasons and regions, the interactions between these variables and GPP-1M vary between Southern and Northern Europe. We find that GPP-1M deficits play a major role in summer in Southern Europe, while the preconditions of summer wildfires in Northern Europe are described by low SPEI-3M and soil moisture deficits along with GPP-1M surpluses (see Figs. 2 and 3). This points to the increasing likelihood of wildfire occurrence under warmer climate conditions as prolonged drought conditions create flammable conditions in all regions and seasons of Europe (see Figs. 3 and 4 and Supplementary Material Fig. S2), as exemplified in the 2018 regional fires that occurred during drought conditions in Central and Northern Europe[10].

Wildfire occurrence is often the result of moderate but compounding anomalies of complex hydro-meteorological and land-surface drivers (see Fig. 2). Therefore, it is highly prone to changes in climate that affect temperature, wind speed, precipitation deficits, and VPD. Results from studies relying on future projections of the FWI show that fire danger and season length increase significantly in various regions of Europe[15,17,61,62]. These increases, as well as observations of wildfire occurrences outside of the typical wildfire season, such as winter wildfires in colder climates—i.e., Scandinavia, Eastern Europe, and the mountain regions of the Alps—must be considered in developing wildfire preparedness measures and planning of wildfire fighting resources. In the summer, wildfire awareness and preparedness is generally high, while the danger for wildfires in winter is more variable as wildfires in this season are driven by conditions that develop on sub-monthly timescales (see Fig. 4c). Our results suggest to consider wildfire risk in all seasons and regions of Europe, and to implement suitable adaptation measures that range from creating awareness and wildfire targeted forest management to investing in additional firefighting resources[53].

In summary, wildfire drivers are manifoldly affected by changing climate conditions. The likelihood of wildfires in each season and region in Europe is promoted by different combinations of emerging hydro-meteorological and land-surface drivers over multiple timescales. Our findings illustrate the regional and seasonal key drivers of wildfires and reveal their multivariate nature. Many of these drivers will be amplified under climate change, which increases wildfire risk and therefore calls for the development and implementation of effective measures to prevent and prepare for wildfires.

Methods

Study region

In this study, we focus on the European continent, which is characterized by an arid climate with dry summers in the South, a temperate climate with warm summers in the central and western parts, a cold climate with warm and dry summers in the East, and a cold climate in the North[12]. We split our study area into eight different climate regions: British Isles (BI), Scandinavia (SC), France (FR), Mid-Europe (ME), Eastern Europe (EA), Alps (AL), Iberian Peninsula (IP), and Mediterranean (MD) following the PRUDENCE project[37], which accounts for the north-south and east-west climate gradients of Europe. A regionalization just considering a north-south gradient, such as the SREX regions[63], would not reflect the transition from an Atlantic-influenced to a continentally influenced climate within the study domain.

Most of the PRUDENCE regions encompass mountain ranges, which are climatologically different from the rest of the region. Therefore, we specifically analyze the wildfire seasonality and drivers for mountain and non-mountain regions. To distinguish between mountain and non-mountain regions, we use the Global Mountains K3 dataset of ref. 64, because it was specifically developed to study global ecosystems. The Global Mountains K3 dataset was derived from the Global multi-resolution terrain elevation (GMTED2010)[65] dataset at 250 m resolution[64], and differentiates between high, scattered high, low, and scattered low mountains. Here, we combine high and scattered high mountains to derive a mask of high mountain regions (see "FireCCI51 and derived wildfire events" under the "Methods" section).

FireCCI51 and derived wildfire events

We use the daily burned area from ESA FireCCI version 5.1 (FireCCI51) based on MODIS for the time period 2001–2020[39]. The algorithm of the FireCCI51 product is based on a hybrid approach of active wildfire and burned area detection[39]. The thermal information of active wildfire detections is used to derive the start dates of fires, and change detection is used to classify the burn status of a pixel[39].

We derive wildfire events for our analysis based on the daily pixel products of ESA's FireCCI51 product at 250 m. For each pixel and day in the 20-year time period, we filter for natural vegetation land cover classes that do not include croplands or flooded areas at any fraction (see land cover categories 50–150 in Annex 1 in ref. 66). Then, we apply a filter on the detection confidence and select only wildfire detections with a confidence level larger than 0.6 to reduce potential false-positive wildfire detections from reflectance-based confusion due to sunglint[57,67], cloud shadows, or angular effects[51]. We then create daily binary masks of wildfire (non-) occurrences, where 1 marks wildfire pixels and 0 marks non-wildfire pixels. These daily binary masks are resampled to approximately 5.5 km resolution, which represents the grid cell size of the reanalysis dataset CERRA (see section "Hydro-meteorological and land-surface drivers" under the "Methods" section), by using the count of burned pixels on a given day. After resampling, we remap the coarsened 5.5 km grid from geographical coordinates to the projection of the CERRA dataset, i.e., the Lambert Conformal Conic projection, by applying a first-order conservative remapping[68] (CDO version 1.9.6[69]). Based on the number of originally burned pixels, we derive burned area in hectares by multiplying the number of burned pixels by the original pixel area at the given latitude for the 0.0022457331 degree grid cell resolution of the FireCCI51 pixel product[66], e.g., approximately 250×250 m = 62500 m^2 at the equator. Similar to the FireCCI51 grid product at 0.25 degree (30 km) resolution (see FireCCI51 Product User Guide[66]), we assume that an original pixel is 100% burned when aggregating the original grid cells to the CERRA resolution of 5.5 km.

From this 5.5 km daily observation-based product, we identify wildfire events in two steps and pixel-by-pixel: First, we drop all wildfire pixels that belong to fires smaller than 40 ha because detecting small fires is challenging

and uncertain[39]. We use a 40 ha threshold as a compromise between threshold values suggested in the studies of refs. 40,45,52,70, [52] explain that, depending on the satellite overpass angle, the minimum detectable wildfire size of active fires increases by a factor of three (i.e., 39 ha for the MODIS active fire product[52]). While the minimum fire size thresholds suggested in refs. 40,45,70 vary between 21 and 100 ha, the FireCCI51 product has a higher resolution than the MODIS active fire product, allowing us to use a smaller threshold. We therefore use a threshold of 40 ha for minimum fire size, which compromises between detection uncertainties and ensuring that we analyze wildfires of impact. Second, we identify the start dates of wildfire events in each pixel by applying a right-aligned rolling window of 4 days that derives the cumulative sum of burned area over the time window. We use a 4-day time window to avoid the fragmentation of large fires into multiple events and to merge smaller fires into one event. This 4-day threshold is consistent with the original FireCCI51 algorithm and has been shown to be a good compromise between shorter time steps that would lead to more fragmented fire events and longer time steps that would lead to fewer but larger fire events[71]. The rolling cumulative burned area indicates whether a wildfire is active for multiple days. As long as the cumulative sum of burned area in the right-aligned rolling window is larger than zero, the last day of the wildfire is still within the considered time window. Whenever the rolling sum in a pixel reaches zero again, we subtract 3 days to identify the end date of the wildfire event to account for the 3 additional days in the rolling window. Third, based on the identified start and end dates, we derive the total burned area of each wildfire event in each pixel as the sum of burned area between the start and end dates. Fourth, we assign an event-label to each wildfire event in each pixel, based on the start date and location, represented by the 5.5 km cell's centroid. Lastly, we split this dataset into (a) days on which wildfire events start, (b) days on which wildfires burn, including the start date, and (c) days with no wildfires starting or burning, which corresponds to all days not included in (b). In our analyses and presented results, we use subset (a) to describe conditions on the days when wildfires start and subset (c) for conditions on non-wildfire days.

Hydro-meteorological and land-surface drivers

For the analysis of hydro-meteorological and land-surface drivers of wildfires, we use three data products: (1) the Copernicus European Regional Reanalysis dataset (CERRA)[72] for daily temperature (mean and maximum; 2-m), 10-m surface wind speed, relative humidity (2-m), evaporation, snow depth, volumetric soil moisture, and radiation, (2) the CERRA-Land dataset[73] for daily precipitation, and (3) the MODIS' MOD17A2H product[74] for GPP, which is available at a temporal resolution of 8 days. To make the values and value ranges of all variables comparable with each other, we standardize all variables to unit variance (mean of 0 and SD of 1 following a normal distribution) by using their empirical distribution function (see "Methods"—"Standardization of wildfire driving variables" section). We provide an overview of the variables, their data sources, and their standardization method in Table 1.

CERRA and CERRA-Land are high-resolution (5.5 km), deterministic reanalysis products that cover the domain of Europe for the time period 1984–2021 at 3-hourly temporal resolution[75] and are driven by lateral boundary conditions from ERA-5[72,76]. We derive precipitation from CERRA-Land because it uses additional quality-controlled precipitation observations in comparison to CERRA. CERRA and CERRA-Land show a better representation of most climate variables at surface levels and also capture local extreme events better than ERA-5[75,77]. Based on the original variables from CERRA, we derive VPD[78], potential evapotranspiration (PET)[78], and the three-month Standardized Evapotranspiration Precipitation Index (SPEI-3M)[79], which are described in detail in the supplementary material chapter Supplementary Methods.

We add GPP data from MODIS (i.e., MOD17A2H v006 product[74]) to our variable collection to account for fuel availability and interactions between hydro-meteorological and land-surface conditions. GPP describes the sum of daily photosynthesis[74] and is used as an indicator for available biomass to burn in this study. The MOD17A2H product estimates GPP by combining the measured fraction of absorbed photosynthetically active radiation (FPAR) with additional data sources for photosynthetically active radiation (PAR), i.e., meteorological fields from GMAO/NASA[80], and biome-specific parameters for radiation use efficiency, which describes how efficiently absorbed radiation is converted to vegetation productivity[74]. The product is available as 8-day composites at 500 m spatial resolution, which represent cumulative GPP for the 8-day period[74]. To match the spatial resolution of the GPP product with that of the other driver variables, we resample and regrid the 500 m MODIS grid to the CERRA reanalysis resolution and projection in the same manner as we processed the FireCCI51 250 m dataset (for details, see "FireCCI51 and derived wildfire events" under the "Methods" section). To account for fluctuations in fuel availability, i.e., available biomass to burn, we apply a rolling window of 4 to the 8-day cumulative GPP from the MOD17A2H product to capture monthly GPP summations (GPP-1M). This rolling window summarizes four times 8-day GPP summations and therefore, represents cumulative GPP for 32 days, which we consider to reflect one month at each 8-day observation point. While GPP-1M is not a direct measure of fuel load, monthly sums allow us to capture moisture losses in biomass, which are relevant for fire activity[26,81,82]. To match the 8-day temporal resolution of the GPP-1M dataset with the daily resolution of the other variables, we apply a linear interpolation to fill in the days between the 8-day GPP-1M values, after the standardization (see "FireCCI51 and derived wildfire events" under the "Methods" section).

Spatial and temporal patterns of wildfires

We evaluate spatial and temporal patterns of wildfires for each region, its respective mountain and non-mountain regions, and for four different seasons, i.e., spring (MAM), summer (JJA), fall (SON), and winter (DJF). To study spatial patterns, we derive the cumulative burned area between 2001 and 2020 of each CERRA grid cell. To investigate seasonal patterns, we derive the seasonal fraction of total annual burned area and number of wildfires separately for the mountain and non-mountain regions in each region.

Standardization of wildfire driving variables

We standardize all driver variables, i.e., maximum temperature, vapor pressure deficit (VPD), 10-m surface wind speed, GPP, precipitation, soil moisture, and snow depth, to unit variance using their empirical distribution[83]. We do not standardize the SPEI-3M because it is already a standardized index[79]. Through the standardization, each daily value of the 20-year time series (2001–2020) is expressed as an SD from the climate mean, which equals zero. We apply the standardization pixel-by-pixel for each CERRA grid cell to account for local climate conditions and to be able to compare the results across regions.

The standardization based on the empirical distribution is applied in two steps: First, we estimate the empirical cumulative distribution function $F(x)$ by estimating the percentile rank of each value $P(x)$. The probabilities in $F(x)$ and the respective percentile scores $P(x)$ are based on the rank given by the indicator function $\mathbb{1}\ (X_i \le x)$. The indicator function describes the estimation of the rank by being equal to 1 if $X_i \le x$ and equal to 0 if $X_i \ge x$ for the given number of observations (n):

$$P(x) = F(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}(X_i \le x). \quad (1)$$

For variables, which have minimum values of zero on the left side of their distribution (i.e., VPD, precipitation, snow depth; see Table 1), we randomize the variables' zero values with a very small μ (i.e., 0.0e-27) following a normal distribution. The goal of this step is to avoid the over-representation of zeros in the value ranking. Second, we map the percentile scores $P(x)$ to the quantiles of a normal distribution $z(x)$, by using the inverse

Table 1 | Variables used to derive the hydro-meteorological and land-surface drivers of wildfires in the analysis

Variable	Unit	Temporal resolution	Aggregation method	Standardization method	Data source
Temperature (mean)	K	Daily	Mean	Empirical	CERRA
Temperature (max)*	K	Daily	Max	Empirical	CERRA
Relative humidity	%	Daily	Mean	Empirical	CERRA
VPD*	kPa	Daily	-	Empirical with randomized zeros	CERRA
Total precipitation	mm	Daily	Sum	Empirical with randomized zeros	CERRA-Land
Volumetric soil moisture*	mm	Daily	Mean	Empirical	CERRA
10-m Surface wind speed*	m/s	Daily	Mean	Empirical	CERRA
Snow depth*	m	daily	Mean	Empirical with randomized zeros	CERRA
GPP*	kg C/m^2	8-days	-	Empirical	MODIS MOD17A2Hv006
Evaporation	mm	Daily	Sum	Empirical	CERRA
Potential Evapotranspiration	mm	Daily	Sum	Empirical	CERRA
Surface thermal radiation downwards	J m^{-2}	Daily	Sum	-	CERRA
Surface solar radiation downwards	J m^{-2}	Daily	Sum	-	CERRA
Surface net thermal radiation	J m^{-2}	Daily	Sum	-	CERRA
Surface net solar radiation	J m^{-2}	Daily	Sum	-	CERRA
3-month Standardized precipitation-evaporation index (SPEI-3M)*	SD	Daily	-	None	CERRA

Final variables used as drivers in the analysis are marked with an asterisk (*), the remaining variables were used for preliminary analyses only.

of the cumulative distribution function (Φ^{-1}):

$$z(x) = \Phi^{-1}(P(x)). \quad (2)$$

We evaluate the standardized wildfire driving variables for the start date of wildfire events and compare them with the conditions of non-wildfire days. To derive two distinct datasets, we sample the drivers for the start dates of wildfires and days when no wildfires occur, i.e., datasets (a) and (c) described in "FireCCI51 and derived wildfire events" under the "Methods" section. Based on these two samples, we test for independence (p-value ≤ 0.05) using the non-parametric Mann–Whitney U-test[84] to show significant differences in hydro-meteorological and land-surface pre-conditions between wildfire start days in comparison to non-wildfire days.

Variable importance of wildfire drivers

We use random forest models[85] to quantify the importance of different hydro-meteorological and land-surface drivers for the occurrence of wildfire events. Random forest models represent a supervised machine learning algorithm, which creates multiple decision trees by randomly sampling training observations for a classification problem (in this case, wildfire vs non-wildfire events). The resulting model represents the average of all decision trees and calculates the variable importance based on the mean decrease in impurity for the different tree splits, also known as Gini importance[85].

Here, we set up multiple random forest models for each region and season using maximum temperature, VPD, 10-m surface wind speed, GPP (cumulative over 32 days), the SPEI-3M, soil moisture, and snow depth as predictor variables and the binary flag of a wildfire event start as a target variable. We use the Python package SCIKIT-LEARN V.1.3.0[86] and its default hyper-parameters (see https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html), as hyper-parameter tuning does not necessarily improve prediction skill[87]. First, we randomly select non-wildfire days in the respective season and region that match the number of observed wildfire days to account for dataset imbalance. Second, we perform a five-fold cross-validation by training the model on 80% of these samples and evaluating it on the remaining 20%. To capture the variability of the random-forest decision trees, we repeat the previous steps 100 times. The presented results on feature importance show the mean importance for each region and season across the 100 random forest models.

We evaluate model performance using the overall accuracy of the confusion matrix, which shows the ratio between correct classifications (true positive and true negative) and all classifications[88]. The mean over all accuracies for 100 random forest models in each region and season ranges between 0.76 and 0.96 (see Supplementary Material Fig. S2). We observe the highest accuracies in winter in all regions, except the Mediterranean and the Iberian Peninsula. In these two regions, we also observe the overall lowest accuracies, which occur in summer and fall, when conditions are strongly wildfire-favoring, also on days with no wildfire occurrences. Given the high overall accuracies (see Supplementary Material Fig. S5), we expect to derive meaningful findings in terms of the variable importance of wildfire drivers.

Time of precondition emergence of wildfire drivers

We study how long wildfire preconditions have been anomalous before wildfire events, by calculating the rolling mean of the standardized driving variables for different time steps. We increase the window sizes of the right-aligned rolling mean iteratively from daily levels (i.e., 0 (day of wildfire-start), 3, 7, 14 days before wildfires ignite) to monthly levels (1–12 months, where we count 30 days for each month) up to one year (i.e., 360 days) before the start day of each wildfire event for all variables. We assume that the incrementally increasing window size indicates how long anomalous conditions of wildfire driving preconditions persist prior to a wildfire event. To set this information in context with non-wildfire conditions, we derive the same rolling mean for all days on which no wildfires occur (see non-wildfire dataset (c) in "FireCCI51 and derived wildfire events" under "Methods" section). We conduct the precondition analysis on the wildfire event scale and then average the results for each region and season. This results in two graphs, one for preconditions on wildfire start days and one for those on non-wildfire days (see Fig. 4a), for each region, season, and variable. In the next step, we identify the time point where these two graphs deviate from each other. Specifically, we derive the difference between the graph for

wildfire start days and the graph for non-wildfire days and use different SD levels, i.e., 1 SD, 1.5 SD, and 2 SD, of this difference as a buffer for natural variability on the non-wildfire graph. Last, we identify the time step, where the anomalies of the wildfire days emerge from these SD buffers around the non-wildfire anomalies, and call this time step "time of precondition emergence" (ToPE). We find that the 1 SD threshold coincides consistently with the onset of the divergence between wildfire and non-wildfire preconditions across variables and regions (see Supplementary Material Fig. S4). Therefore, we use ToPE based on one SD in our analysis. We use +1 SD for maximum temperature, VPD, and 10-m surface wind speed, and −1 SD for SPEI-3M, soil moisture, and snow depth. For GPP (GPP-1M), we find that in spring and summer GPP-1M is more abundant on non-wildfire days, whereas in fall and winter GPP-1M is more abundant on wildfire-days compared to non-wildfire days (see Fig. 2), which is why we derive the lower level of ToPE (−1 SD) for spring and summer and the upper anomaly persistence (+1 SD) for fall and winter.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

FireCCI data can be downloaded from the Copernicus repository https://doi.org/10.24381/cds.f333cf85. CERRA data can be downloaded from the Copernicus repository https://doi.org/10.24381/cds.622a565a. CERRA-Land data can be downloaded from the Copernicus repository https://doi.org/10.24381/cds.a7f3cd0b. MODIS Gross Primary Productivity can be downloaded from https://www.earthdata.nasa.gov/data/catalog/lpcloud-mod17a2h-061/. The mountain region mask is available at https://data.usgs.gov/datacatalog/data/USGS:638fbf72d34ed907bf7d3080. We provide the derived wildfire events and drivers, which we used in the analysis, through a dataset repository on Envidat (https://doi.org/10.16904/envidat.575).

Code availability

The code to reproduce the analysis and the figures is available on GitLab: (https://gitlab.com/jumi26/wildfire-drivers-in-europe.git).

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Author contributions

J.M., D.T., and M.I.B. conceptualized the study. J.M. conducted the data collection, analysis, and visualization. J.M. wrote the original draft, which was revised and edited by D.T., M.I.B., and J.M.

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